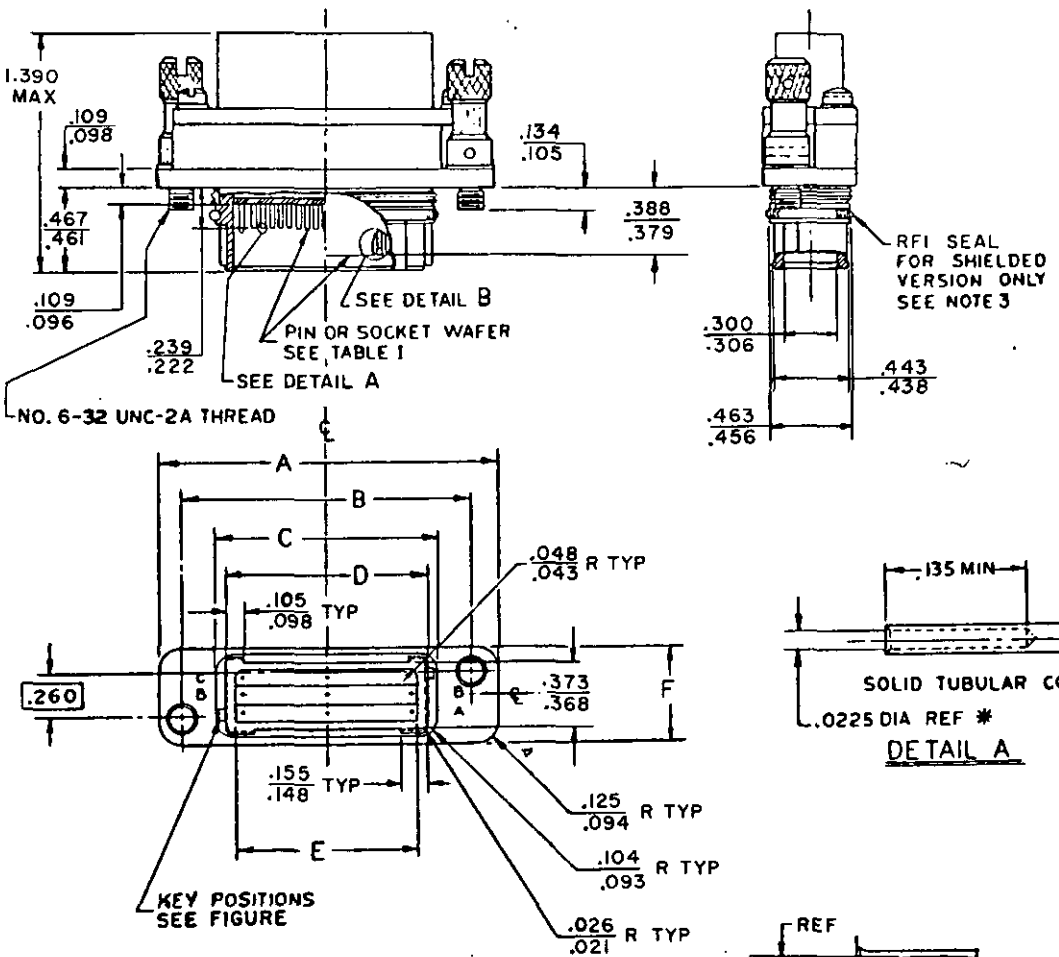
MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, PLUG, ENVIRONMENT RESISTANT, FOR USE WITH FLEXIBLE FLAT CONDUCTOR CABLE AND ROUND WIRE, REMOVABLE WAFERS (POTTED), FLAT, BACKSHELL

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-55544.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

NOT FOR AIR FORCE USE UNLESS APPROVED BY THE PROCURING ACTIVITY



* SPRING MEMBER IS ON CONTACT PIN. CONTACT PIN DIAMETER .025 MAY BE VARIED TO PROVIDE A MAX ENGAGING FORCE OF 6 OZ AND A MIN SEPARATION FORCE OF 1/2 OZ.

FSC 5935

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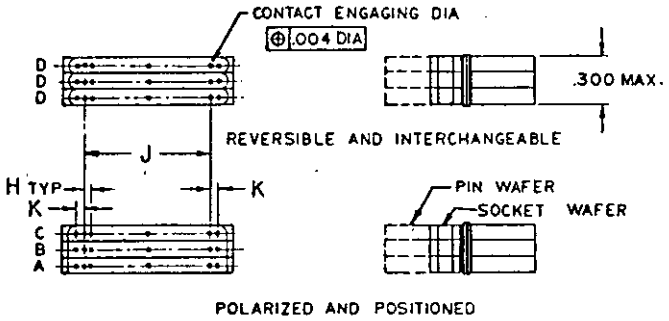
TABLE I. Shell dimensions.

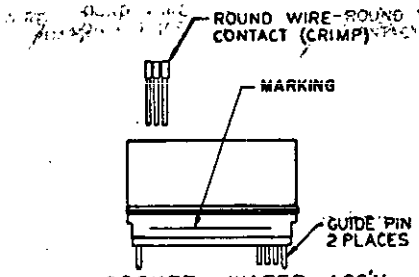
Military part no.	A	B Basic	C	D	E	F
M55544/17-**B3*						
M55544/17-**C3*	1.906 (48.41) 1.895 (48.13)	1.625 (41.28)	1.233 (31.33) 1.228 (31.19)	1.113 (28.27) 1.108 (28.14)	1.032 (26.21) 1.027 (26.09)	.546 (13.87) .535 (13.59)
M55544/17-**D3*						
M55544/17-**E3*						
M55544/17-**F3*	3.505 (89.03) 3.480 (88.39)	3.188 (80.98)	2.733 (69.42) 2.728 (69.29)	2.613 (66.37) 2.608 (66.24)	2.535 (64.39) 2.530 (64.26)	.559 (14.20) .548 (13.92)
M55544/17-**G3*						

INCHES	MM	INCHES	MM
.019	.48	.148	3.76
.021	.53	.155	3.94
.0225	.57	.222	5.64
.025	.64	.239	6.07
.026	.66	.260	6.60
.037	.94	.300	7.62
.043	1.09	.306	7.77
.048	1.22	.368	9.35
.093	2.36	.373	9.47
.094	2.39	.379	9.63
.096	2.44	.388	9.86
.098	2.49	.438	11.13
.104	2.64	.443	11.25
.105	2.67	.456	11.58
.109	2.77	.461	11.71
.125	3.18	.463	11.76
.134	3.40	.467	11.86
.135	3.43	1.390	35.31

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WAFER ARRANGEMENT (SEE TABLE II)





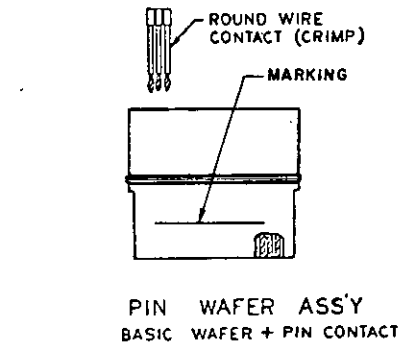


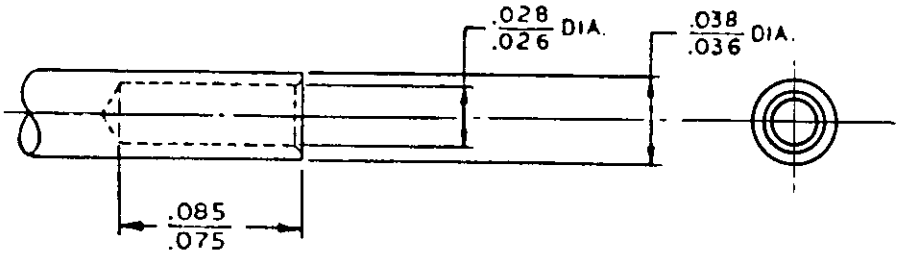
TABLE II. Wafer arrangement.

Cable width code	Cable width	Contact spacing no.	No. of contacts per row	H	J	K	Total contacts per no. of cable layers			Max. mating force lbs	Max. unmating force lbs
							1	2	3	shell only	shell only
B	.500	2	7	.050	.300	.0875	7	14	21	6	2
	.500	3	6	.075	.375	.0500	6	12	18		
	.500	4	4	.100	.300	.0875	4	8	12		
C	1.000	2	17	.050	.800	.0625	17	34	51	8	4
	1.000	3	12	.075	.825	.0500	12	24	36		
	1.000	4	9	.100	.800	.0625	9	18	27		
D	1.500	2	27	.050	1.300	.0625	27	54	81	10	6
	1.500	3	18	.075	1.275	.0750	18	36	54		
	1.500	4	14	.100	1.300	.0625	14	28	42		
E	2.000	2	37	.050	1.800	.0625	37	74	111	12	8
	2.000	3	25	.075	1.800	.0625	25	50	75		
	2.000	4	19	.100	1.800	.0625	19	38	57		
F	2.500	2	47	.050	2.300	.0625	47	94	141	14	10
	2.500	3	32	.075	2.325	.0500	32	64	96		
	2.500	4	24	.100	2.300	.0625	24	48	72		
G	3.000	2	57	.050	2.800	.0625	57	114	171	16	12
	3.000	3	38	.075	2.775	.0750	38	76	114		
	3.000	4	29	.100	2.800	.0625	29	58	87		

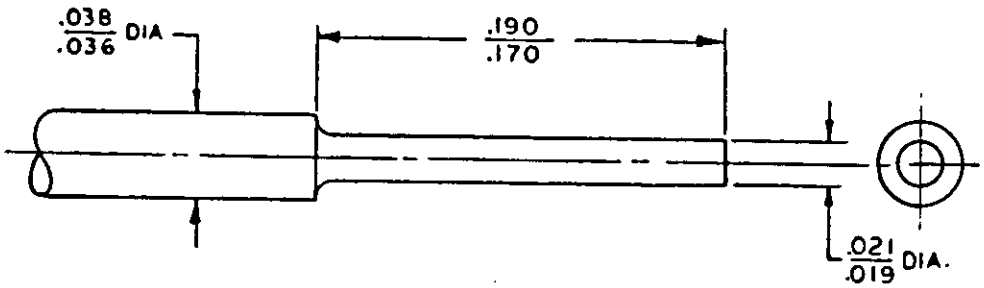
INCHES	MM
.004	.10
.0500	1.27
.0625	1.59
.0750	1.91
.0875	2.22
.100	2.54
.300	7.62
.375	9.53
.500	12.70
.800	20.32
.825	20.96
1.000	25.40
1.275	32.39
1.300	33.02
1.500	38.10
1.800	45.72
2.000	50.80
2.300	58.42
2.500	63.50
2.775	70.49
2.800	71.12
3.000	76.20

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CONTACT TERMINATION DESIGN DATA



CRIMP TYPE CONTACT H,K,M,R



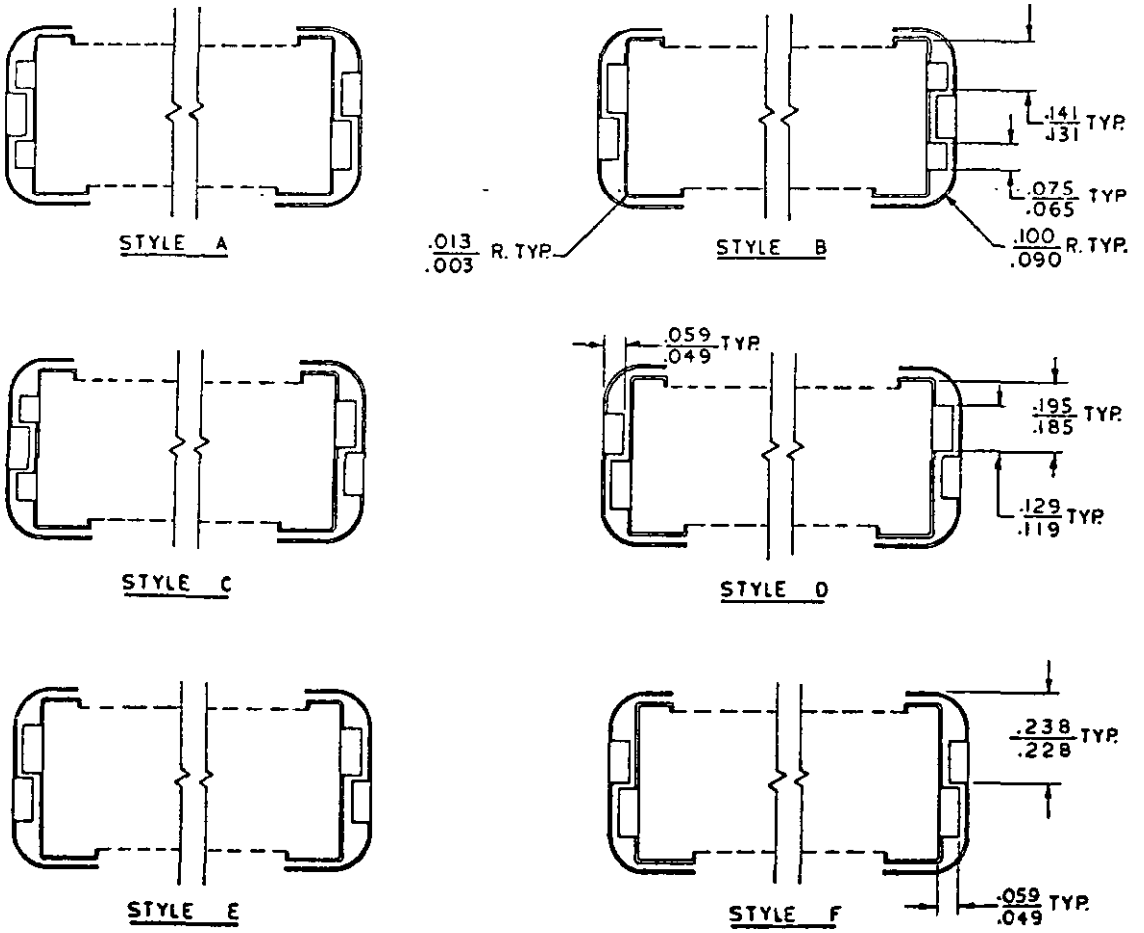
WELD OR LAP SOLDER TYPE CONTACT G,J,L,N

INCHES	MM
.019	.48
.021	.53
.026	.66
.028	.71
.036	.91
.038	.97
.075	1.91
.085	2.16
.170	4.32
.190	4.83

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KEYING POSITIONS

FRONT VIEW OF PLUG FACE SHOWN, RECEPTACLE KEYS SHOWN IN MATED CONDITION.

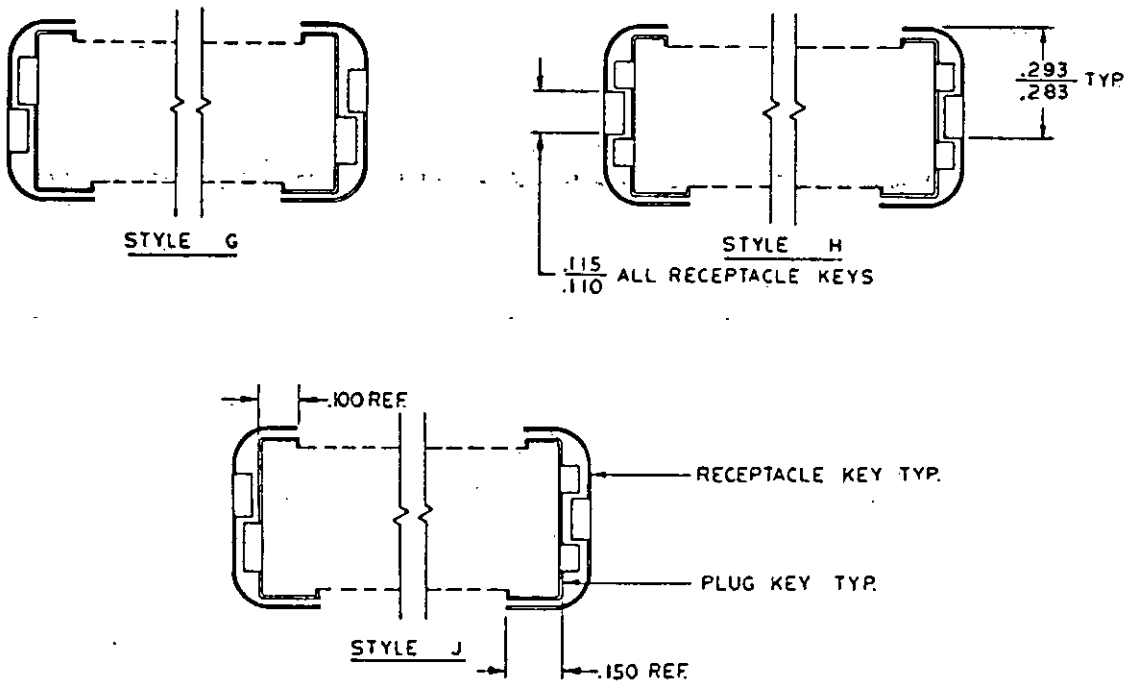


INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.090	2.29	.141	3.58
.013	.33	.100	2.54	.185	4.70
.049	1.24	.119	3.02	.195	4.95
.059	1.50	.129	3.28	.228	5.79
.065	1.65	.131	3.33	.238	6.05
.075	1.91				

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Keying positions (continued)

Front view of plug face shown, receptacle keys shown in mated condition.



INCHES	MM
.100	2.54
.110	2.79
.115	2.92
.150	3.81
.283	7.19
.293	7.44

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. This connector design is for a three cable layer model only.

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REQUIREMENTS:

Modular wafer connector: Pin or socket wafers will fit either plug or receptacle shells. Wafers are available polarized and positioned or reversible and interchangeable.

Contacts: Flat cable inserted prior to welding. Round wires inserted after crimping.

Contact material: Copper alloy.

Cable seal method: Potting of the individual modular wafer.

Shell material: Aluminum alloy.

Mating and unmating forces: Add 8 ounces per contact for mating and 6 ounces per contact for unmating to shell forces shown in table II.

Contact category and wafer description: See table III.

Contact resistance: The maximum voltage drop, measured at the wire at the rear of the contact is 24 millivolts at 3 amperes.

Low level contact resistance: The maximum voltage drop, measured at the wire at the rear of the contact is $.24 \times 10^{-6}$ volts at 10^{-6} amperes.

TABLE III. Contact category and wafer description.

Type	Contact category and wafer description
G	Pin, polarized and positioned, weld (flat cable)
H	Pin, polarized and positioned, crimp (round wire)
J	Pin, reversible and interchangeable, weld
K	Pin, reversible and interchangeable, crimp
L	Socket, polarized and positioned, weld
M	Socket, polarized and positioned, crimp
N	Socket, reversible and interchangeable, weld
R	Socket, reversible and interchangeable, crimp

Cable layers and shielding: See table IV.

TABLE IV. Number of cable layers and shielding.

Code number	Type of cable
3	Three cable layers, unshielded
6	Three cable layers, shielded

Mating characteristic: This connector is designed to be mated with receptacle having compatible contacts, M55544/18, M55544/20, and M55544/22.

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Part number: Consists of the basic document number of this specification sheet, and a coded dash number as follows:

Example:

M55544/17-	G	2	C	3	A
Specification sheet number See table I	Contact category See table IV	Contact spacing number See table II	Cable width code See table II	No. of cable layers and shielding See drawing note 3 and table IV	Keying position See figure

Custodians:
Army - EL
Navy - AS
Air Force - 11

Review activities:
Army - MU, MI, AV, EL, SL
Navy - EC, MC, AS
Air Force - 15, 17, 80
DSA - ES
NASA
NS

User activities:
Army - ME, WC
Navy -
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1703-17)

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